

Enterprise Access Point & Centralized DHCP Relay Simulation

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Tool Used:

Cisco Packet Tracer

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PROJECT LABORATORY OVERVIEW

This Cisco Packet Tracer simulation is designed to showcase hands-on proficiency in enterprise network infrastructure, Layer 2/3 traffic management, and practical network troubleshooting.

The primary objective of this laboratory is to simulate centralized IP address distribution across multiple segmented VLANs using a single central DHCP Server via IP helper-address (DHCP Relay) over an enterprise MDF/IDF switch hierarchy.

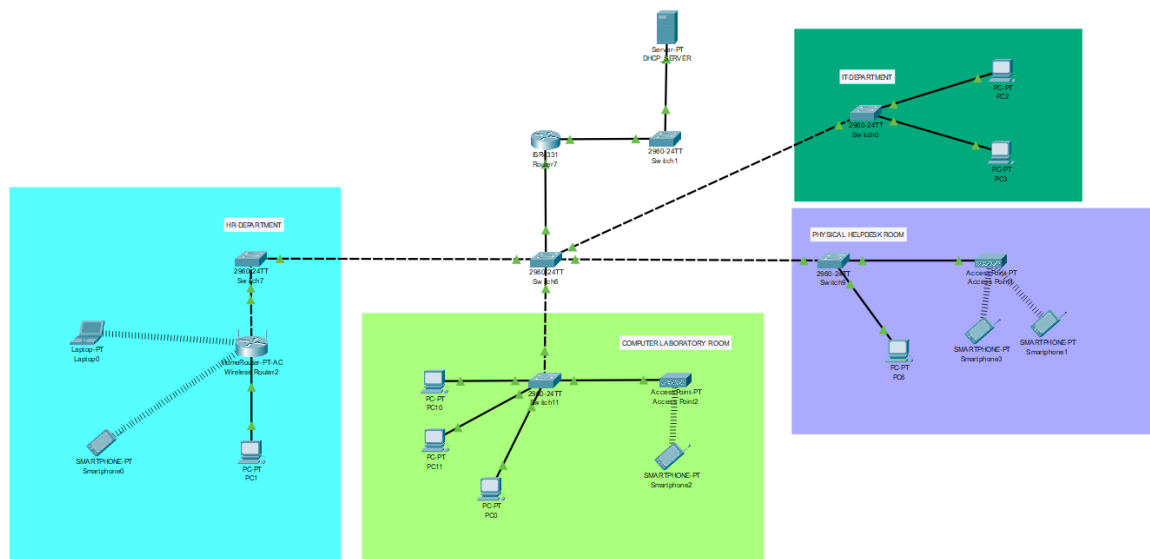


Figure 1: DHCP Relay Topology

The topology models a multi-department enterprise network using a standard Main Distribution Frame (MDF) and Intermediate Distribution Frame (IDF) hierarchy. It also demonstrates core networking concepts such as VLAN segmentation, VLSM/CIDR Subnetting, inter-VLAN routing, DHCP, layer 2 security mitigation, and real-world problem scenarios (break/fix).

NETWORK ARCHITECTURE AND HARDWARE ROLES

The topology models a multi-department enterprise network using a standard Main Distribution Frame (MDF) and Intermediate Distribution Frame (IDF) hierarchy.

Device Name	Device Model / Type	Network Role & Function
MAIN_ROUTER	Cisco ISR 4331/K9	Central Layer 3 Gateway performing Router-on-a-Stick (ROAS) Inter-VLAN routing and acting as the DHCP Relay Agent via IP Helper-Address
BRIDGE_SWITCH_1	Cisco Switch	Layer 2 bridge link connecting MAIN_ROUTER directly to the central DHCP Server (Server0).
MDF_SWITCH (Switch 6)	Cisco 2960 Switch	Main Distribution Frame (MDF) core switch managing primary trunk connections across all departments.
HR_SWITCH (Switch 7)	Cisco 2960 Switch	IDF servicing the Human Resources Department (VLAN 20).
COMLAB_SWITCH (Switch 11)	Cisco 2960 Switch	IDF servicing the Computer Laboratory Room (VLAN 10).
HELPDESK_SWITCH (Switch 9)	Cisco 2960 Switch	IDF servicing the Physical Helpdesk Room (VLAN 30).

ITDEPT_SWITCH (Switch 0)	Cisco 2960 Switch	IDF servicing the IT Department (VLAN 50).
Guest AP (AP 1)	Lightweight Access Point	Wireless Access Point broadcasting Guest Wi-Fi (VLAN 40).
ComLab AP (AP 2)	Lightweight Access Point	Wireless Access Point broadcasting Computer Lab Wi-Fi (VLAN 10).

IP SUBNETTING CALCULATIONS

The laboratory consists of multiple department VLANs, their planned capacities, and the allocated CIDR block size for each department.

Given Base Network: 192.168.10.0/24

Department / VLAN	Planned Capacity	Allocated CIDR Block Size
VLAN 10 (COMLAB)	25 - 30 Hosts	/27 (30 Hosts)
VLAN 20 (HR_DEPT)	10 - 14 Hosts	/28 (14 Hosts)
VLAN 30 (IT_DEPT)	10 - 14 Hosts	/28 (14 Hosts)
VLAN 40 (GUEST_WIFI)	10 - 14 Hosts	/28 (14 Hosts)
VLAN 50 (SERVERS)	4-6 Hosts	/29 (6 Hosts)
VLAN 100 (UNUSED_PORTS)	–	–

Assumed planned hosts for each department:

- Computer Laboratory Room - 30 hosts
- HR Department - 14 hosts
- IT Department - 14 hosts
- Helpdesk Room (for guest Wi-Fi) - 10 hosts
- Helpdesk Room (company's devices) - 6 hosts
- Servers - 4 hosts

Detailed calculations:**1st subnet/VLAN 10 (computer laboratory room):**

- Network: 192.168.10.0/27
- Hosts: 192.168.10.1 - 192.168.10.30 (/27)
- Broadcast: 192.168.10.31/27

2nd Subnet/VLAN 20 (HR Department):

- Network: 192.168.10.32/28
- Hosts: 192.168.10.33 - 192.168.10.46 (/28)
- Broadcast: 192.168.10.47/28

3rd Subnet/VLAN 30 (IT Department):

- Network: 192.168.10.48/28
- Hosts: 192.168.10.49 - 192.168.10.62 (/28)
- Broadcast: 192.168.10.63/28

4th Subnet/VLAN 40 (Helpdesk Room - guest Wi-Fi):

- Network: 192.168.10.64/28
- Hosts: 192.168.10.65 - 192.168.10.78 (/28)
- Broadcast: 192.168.10.79/28

5th Subnet/VLAN 50 (Server):

- Network: 192.168.10.80/29
- Hosts: 192.168.10.81 - 192.168.10.86 (/29)
- Broadcast: 192.168.10.87/29

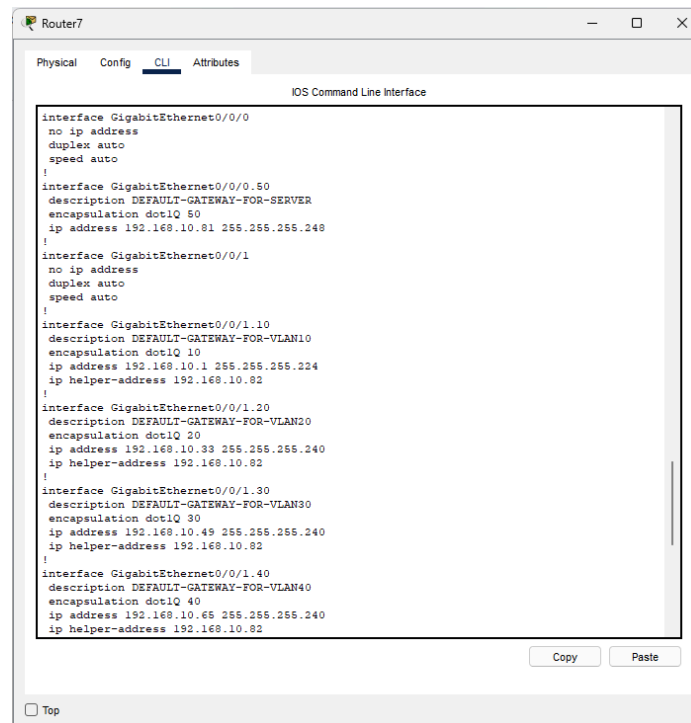
CONFIGURATIONS

This section aims to showcase and demonstrate the necessary configurations made throughout this entire laboratory setup.

MAIN ROUTER CONFIGURATION

1. Cisco ISR4331/K9 configuration (Router 7 - MAIN_ROUTER):

- **Purpose:** Acting as a centralized router performing inter-VLAN routing, and relaying IP requests via IP helper-address from the DHCP server to the end devices and vice versa.
- **Configurations:**
 - Configured GigabitEthernet0/0/0.50 interface as VLAN 50, serving as a default gateway pathway for the server.
 - On the other hand, GigabitEthernet0/0/1 is used as an interface for VLANs 10, 20, 30, and 40.
- **Running-configuration Screenshot:**



```
Router7
Physical Config CLI Attributes
IOS Command Line Interface

interface GigabitEthernet0/0/0
no ip address
duplex auto
speed auto
!
interface GigabitEthernet0/0/0.50
description DEFAULT-GATEWAY-FOR-SERVER
encapsulation dot1Q 50
ip address 192.168.10.81 255.255.255.240
!
interface GigabitEthernet0/0/1
no ip address
duplex auto
speed auto
!
interface GigabitEthernet0/0/1.10
description DEFAULT-GATEWAY-FOR-VLAN10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.224
ip helper-address 192.168.10.82
!
interface GigabitEthernet0/0/1.20
description DEFAULT-GATEWAY-FOR-VLAN20
encapsulation dot1Q 20
ip address 192.168.10.33 255.255.255.240
ip helper-address 192.168.10.82
!
interface GigabitEthernet0/0/1.30
description DEFAULT-GATEWAY-FOR-VLAN30
encapsulation dot1Q 30
ip address 192.168.10.49 255.255.255.240
ip helper-address 192.168.10.82
!
interface GigabitEthernet0/0/1.40
description DEFAULT-GATEWAY-FOR-VLAN40
encapsulation dot1Q 40
ip address 192.168.10.65 255.255.255.240
ip helper-address 192.168.10.82
```

SWITCH CONFIGURATION

1. Configurations created for each switch:

- Created 6 VLANs:
 - VLAN 10 - For computer laboratory
 - VLAN 20 - For HR Department
 - VLAN 30 - For IT Department
 - VLAN 40 - For Guest Wi-Fi
 - VLAN 50 - For Servers
 - VLAN 100 - For unused ports

2. Configuration for switch 6 (MDF_SWITCH):

- **Purpose:** Acting as the main distribution frame for all the switches.
 - **Configurations Created:**
 - The interface range Fa0/1–Fa0/5 is configured as a trunk.
 - Interface range Fa0/6–Fa0/24, along with interface range G0/1–G0/2, is shut down and assigned to the unused port VLAN.
- **Configurations Screenshots:**

```
MDF_SWITCH#sh vlan br
```

VLAN	Name	Status	Ports
1	default	active	
10	COMLAB	active	
20	HR_DEPT	active	
30	IT_DEPT	active	
40	GUEST_WIFI	active	
50	SERVERS	active	
100	UNUSED_PORTS	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
MDF_SWITCH#
```


MDF_SWITCH#sh int tr

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	on	802.1q	trunking	1
Fa0/2	on	802.1q	trunking	1
Fa0/3	on	802.1q	trunking	1
Fa0/4	on	802.1q	trunking	1
Fa0/5	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Fa0/1	1-1005
Fa0/2	1-1005
Fa0/3	1-1005
Fa0/4	1-1005
Fa0/5	1-1005

Port	Vlans allowed and active in management domain
Fa0/1	1,10,20,30,40,50,100
Fa0/2	1,10,20,30,40,50,100
Fa0/3	1,10,20,30,40,50,100
Fa0/4	1,10,20,30,40,50,100
Fa0/5	1,10,20,30,40,50,100

Port	Vlans in spanning tree forwarding state and not pruned
Fa0/1	1,10,20,30,40,50,100
Fa0/2	1,10,20,30,40,50,100
Fa0/3	1,10,20,30,40,50,100
Fa0/4	1,10,20,30,40,50,100
Fa0/5	1,10,20,30,40,50,100

MDF_SWITCH#sh int status

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1		connected	trunk	auto	auto	10/100BaseTX
Fa0/2		connected	trunk	auto	auto	10/100BaseTX
Fa0/3		connected	trunk	auto	auto	10/100BaseTX
Fa0/4		connected	trunk	auto	auto	10/100BaseTX
Fa0/5		connected	trunk	auto	auto	10/100BaseTX
Fa0/6		disabled	100	auto	auto	10/100BaseTX
Fa0/7		disabled	100	auto	auto	10/100BaseTX
Fa0/8		disabled	100	auto	auto	10/100BaseTX
Fa0/9		disabled	100	auto	auto	10/100BaseTX
Fa0/10		disabled	100	auto	auto	10/100BaseTX
Fa0/11		disabled	100	auto	auto	10/100BaseTX
Fa0/12		disabled	100	auto	auto	10/100BaseTX
Fa0/13		disabled	100	auto	auto	10/100BaseTX
Fa0/14		disabled	100	auto	auto	10/100BaseTX
Fa0/15		disabled	100	auto	auto	10/100BaseTX
Fa0/16		disabled	100	auto	auto	10/100BaseTX
Fa0/17		disabled	100	auto	auto	10/100BaseTX
Fa0/18		disabled	100	auto	auto	10/100BaseTX
Fa0/19		disabled	100	auto	auto	10/100BaseTX
Fa0/20		disabled	100	auto	auto	10/100BaseTX
Fa0/21		disabled	100	auto	auto	10/100BaseTX
Fa0/22		disabled	100	auto	auto	10/100BaseTX
Fa0/23		disabled	100	auto	auto	10/100BaseTX
Fa0/24		disabled	100	auto	auto	10/100BaseTX
Gig0/1		disabled	100	auto	auto	10/100BaseTX
Gig0/2		disabled	100	auto	auto	10/100BaseTX

3. Configuration for switch 1 (BRIDGE_SWITCH_1):

- **Purpose:** Acting as the bridge between the Cisco ISR4331/K9 router and the server, and vice versa.
- **Configurations Created:**
 - Int fa0/2 configured as a trunk.
 - Int fa0/1 configured as VLAN 50.
 - Int range Fa0/3 - fa0/24, along with interface range G0/1–G0/2, is shut down and assigned to the unused port VLAN.
- **Configurations Screenshots:**

```
BRIDGE_SWITCH_1#sh vlan br
```

VLAN	Name	Status	Ports
1	default	active	
10	COMLAB	active	
20	HR_DEPT	active	
30	IT_DEPT	active	
40	GUEST_WIFI	active	
50	SERVERS	active	Fa0/1
100	UNUSED_PORTS	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
BRIDGE_SWITCH_1#
```

```
BRIDGE_SWITCH_1#sh int tr
```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/2	on	802.1q	trunking	1
Port	Vlans allowed on trunk			
Fa0/2	1-1005			
Port	Vlans allowed and active in management domain			
Fa0/2	1,10,20,30,40,50,100			
Port	Vlans in spanning tree forwarding state and not pruned			
Fa0/2	1,10,20,30,40,50,100			

```
BRIDGE_SWITCH_1#
```

```
BRIDGE_SWITCH_1#sh int status
Port      Name      Status      Vlan      Duplex  Speed Type
Fa0/1      Name      connected   50        auto    auto  10/100BaseTX
Fa0/2      connected trunk    auto    auto  10/100BaseTX
Fa0/3      disabled 100      auto    auto  10/100BaseTX
Fa0/4      disabled 100      auto    auto  10/100BaseTX
Fa0/5      disabled 100      auto    auto  10/100BaseTX
Fa0/6      disabled 100      auto    auto  10/100BaseTX
Fa0/7      disabled 100      auto    auto  10/100BaseTX
Fa0/8      disabled 100      auto    auto  10/100BaseTX
Fa0/9      disabled 100      auto    auto  10/100BaseTX
Fa0/10     disabled 100      auto    auto  10/100BaseTX
Fa0/11     disabled 100      auto    auto  10/100BaseTX
Fa0/12     disabled 100      auto    auto  10/100BaseTX
Fa0/13     disabled 100      auto    auto  10/100BaseTX
Fa0/14     disabled 100      auto    auto  10/100BaseTX
Fa0/15     disabled 100      auto    auto  10/100BaseTX
Fa0/16     disabled 100      auto    auto  10/100BaseTX
Fa0/17     disabled 100      auto    auto  10/100BaseTX
Fa0/18     disabled 100      auto    auto  10/100BaseTX
Fa0/19     disabled 100      auto    auto  10/100BaseTX
Fa0/20     disabled 100      auto    auto  10/100BaseTX
Fa0/21     disabled 100      auto    auto  10/100BaseTX
Fa0/22     disabled 100      auto    auto  10/100BaseTX
Fa0/23     disabled 100      auto    auto  10/100BaseTX
Fa0/24     disabled 100      auto    auto  10/100BaseTX
Gig0/1     disabled 100      auto    auto  10/100BaseTX
Gig0/2     disabled 100      auto    auto  10/100BaseTX
```

4. Configuration for switch 0 (ITDEPT_SWITCH):

- **Purpose:** Acting as an intermediate distribution frame for the IT Department.
- **Configurations Created:**
 - Int fa0/1 configured as a trunk.
 - Int range fa0/2, fa0/3 configured as VLAN 30.
 - Int range Fa0/4 - fa0/24, along with interface range G0/1–G0/2, is shut down and assigned to the unused port VLAN.
- **Configuration Screenshots:**

```
ITDEPT_SWITCH#sh vlan br
```

VLAN	Name	Status	Ports
1	default	active	
10	COMLAB	active	
20	HR_DEPT	active	
30	IT_DEPT	active	Fa0/2, Fa0/3
40	GUEST_WIFI	active	
50	SERVERS	active	
100	UNUSED_PORTS	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
ITDEPT_SWITCH#
```

```

ITDEPT_SWITCH#sh int tr
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     auto        n-802.1q       trunking     1

Port      Vlans allowed on trunk
Fa0/1     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1,10,20,30,40,50,100

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,20,30,40,50,100

ITDEPT_SWITCH#

```

```

ITDEPT_SWITCH#sh int status
Port      Name      Status      Vlan      Duplex  Speed  Type
Fa0/1     Fa0/1     connected   trunk     auto    auto   10/100BaseTX
Fa0/2     Fa0/2     connected   30        auto    auto   10/100BaseTX
Fa0/3     Fa0/3     connected   30        auto    auto   10/100BaseTX
Fa0/4     Fa0/4     disabled    100       auto    auto   10/100BaseTX
Fa0/5     Fa0/5     disabled    100       auto    auto   10/100BaseTX
Fa0/6     Fa0/6     disabled    100       auto    auto   10/100BaseTX
Fa0/7     Fa0/7     disabled    100       auto    auto   10/100BaseTX
Fa0/8     Fa0/8     disabled    100       auto    auto   10/100BaseTX
Fa0/9     Fa0/9     disabled    100       auto    auto   10/100BaseTX
Fa0/10    Fa0/10    disabled    100       auto    auto   10/100BaseTX
Fa0/11    Fa0/11    disabled    100       auto    auto   10/100BaseTX
Fa0/12    Fa0/12    disabled    100       auto    auto   10/100BaseTX
Fa0/13    Fa0/13    disabled    100       auto    auto   10/100BaseTX
Fa0/14    Fa0/14    disabled    100       auto    auto   10/100BaseTX
Fa0/15    Fa0/15    disabled    100       auto    auto   10/100BaseTX
Fa0/16    Fa0/16    disabled    100       auto    auto   10/100BaseTX
Fa0/17    Fa0/17    disabled    100       auto    auto   10/100BaseTX
Fa0/18    Fa0/18    disabled    100       auto    auto   10/100BaseTX
Fa0/19    Fa0/19    disabled    100       auto    auto   10/100BaseTX
Fa0/20    Fa0/20    disabled    100       auto    auto   10/100BaseTX
Fa0/21    Fa0/21    disabled    100       auto    auto   10/100BaseTX
Fa0/22    Fa0/22    disabled    100       auto    auto   10/100BaseTX
Fa0/23    Fa0/23    disabled    100       auto    auto   10/100BaseTX
Fa0/24    Fa0/24    disabled    100       auto    auto   10/100BaseTX
Gig0/1    Gig0/1    disabled    100       auto    auto   10/100BaseTX
Gig0/2    Gig0/2    disabled    100       auto    auto   10/100BaseTX

ITDEPT_SWITCH#

```

5. Configuration for switch 9 (HELPDESK_SWITCH):

- **Purpose:** Acting as an intermediate distribution frame for the physical helpdesk room.
- **Configurations Created:**
 - Int fa0/1 configured as a trunk.
 - Int fa0/2 configured as VLAN 40.
 - Int fa0/3 configured as VLAN 10 (part of computer laboratory devices).
 - Int range Fa0/4 - fa0/24, along with interface range G0/1–G0/2, is shut down and assigned to the unused port VLAN.
- **Configuration Screenshots:**

```
HELPDESK_SWITCH#sh vlan br
```

VLAN Name	Status	Ports
1 default	active	
10 COMLAB	active	Fa0/3
20 HR_DEPT	active	
30 IT_DEPT	active	
40 GUEST_WIFI	active	Fa0/2
50 SERVERS	active	
100 UNUSED_PORTS	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
HELPDESK_SWITCH#
```

```
HELPDESK_SWITCH#sh int tr
```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Fa0/1	1-1005

Port	Vlans allowed and active in management domain
Fa0/1	1,10,20,30,40,50,100

Port	Vlans in spanning tree forwarding state and not pruned
Fa0/1	1,10,20,30,40,50,100

```
HELPDESK_SWITCH#
```

```
HELPDESK_SWITCH#sh int status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1		connected	trunk	auto	auto	10/100BaseTX
Fa0/2		connected	40	auto	auto	10/100BaseTX
Fa0/3		connected	10	auto	auto	10/100BaseTX
Fa0/4		disabled	100	auto	auto	10/100BaseTX
Fa0/5		disabled	100	auto	auto	10/100BaseTX
Fa0/6		disabled	100	auto	auto	10/100BaseTX
Fa0/7		disabled	100	auto	auto	10/100BaseTX
Fa0/8		disabled	100	auto	auto	10/100BaseTX
Fa0/9		disabled	100	auto	auto	10/100BaseTX
Fa0/10		disabled	100	auto	auto	10/100BaseTX
Fa0/11		disabled	100	auto	auto	10/100BaseTX
Fa0/12		disabled	100	auto	auto	10/100BaseTX
Fa0/13		disabled	100	auto	auto	10/100BaseTX
Fa0/14		disabled	100	auto	auto	10/100BaseTX
Fa0/15		disabled	100	auto	auto	10/100BaseTX
Fa0/16		disabled	100	auto	auto	10/100BaseTX
Fa0/17		disabled	100	auto	auto	10/100BaseTX
Fa0/18		disabled	100	auto	auto	10/100BaseTX
Fa0/19		disabled	100	auto	auto	10/100BaseTX
Fa0/20		disabled	100	auto	auto	10/100BaseTX
Fa0/21		disabled	100	auto	auto	10/100BaseTX
Fa0/22		disabled	100	auto	auto	10/100BaseTX
Fa0/23		disabled	100	auto	auto	10/100BaseTX
Fa0/24		disabled	100	auto	auto	10/100BaseTX
Gig0/1		disabled	100	auto	auto	10/100BaseTX
Gig0/2		disabled	100	auto	auto	10/100BaseTX

```
HELPDESK_SWITCH#
```

6. Configuration for switch 11 (COMLAB_SWITCH):

- **Purpose:** Acting as an intermediate distribution frame for the computer laboratory room.
- **Configurations Created:**
 - Int fa0/1 configured as a trunk.
 - Int range fa0/2, fa0/3, fa0/4, fa0/5 configured as VLAN 10.
 - Int range Fa0/6 - fa0/24, along with interface range G0/1–G0/2, is shut down and assigned to the unused port VLAN.
- **Configuration Screenshots:**

```
COMLAB_SWITCH#sh vlan br
```

VLAN Name	Status	Ports
1 default	active	
10 COMLAB	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5
20 HR_DEPT	active	
30 IT_DEPT	active	
40 GUEST_WIFI	active	
50 SERVERS	active	
100 UNUSED_PORTS	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
COMLAB_SWITCH#
```

```
COMLAB_SWITCH#sh int tr
```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	on	802.1q	trunking	1
Port Vlans allowed on trunk				
Fa0/1	1-1005			
Port Vlans allowed and active in management domain				
Fa0/1	1,10,20,30,40,50,100			
Port Vlans in spanning tree forwarding state and not pruned				
Fa0/1	1,10,20,30,40,50,100			

```
COMLAB_SWITCH#
```

```
COMLAB_SWITCH#sh int status
Port      Name      Status      Vlan      Duplex  Speed Type
Fa0/1     connected trunk       auto      auto  10/100BaseTX
Fa0/2     connected 10          auto      auto  10/100BaseTX
Fa0/3     connected 10          auto      auto  10/100BaseTX
Fa0/4     connected 10          auto      auto  10/100BaseTX
Fa0/5     connected 10          auto      auto  10/100BaseTX
Fa0/6     disabled 100         auto      auto  10/100BaseTX
Fa0/7     disabled 100         auto      auto  10/100BaseTX
Fa0/8     disabled 100         auto      auto  10/100BaseTX
Fa0/9     disabled 100         auto      auto  10/100BaseTX
Fa0/10    disabled 100         auto      auto  10/100BaseTX
Fa0/11    disabled 100         auto      auto  10/100BaseTX
Fa0/12    disabled 100         auto      auto  10/100BaseTX
Fa0/13    disabled 100         auto      auto  10/100BaseTX
Fa0/14    disabled 100         auto      auto  10/100BaseTX
Fa0/15    disabled 100         auto      auto  10/100BaseTX
Fa0/16    disabled 100         auto      auto  10/100BaseTX
Fa0/17    disabled 100         auto      auto  10/100BaseTX
Fa0/18    disabled 100         auto      auto  10/100BaseTX
Fa0/19    disabled 100         auto      auto  10/100BaseTX
Fa0/20    disabled 100         auto      auto  10/100BaseTX
Fa0/21    disabled 100         auto      auto  10/100BaseTX
Fa0/22    disabled 100         auto      auto  10/100BaseTX
Fa0/23    disabled 100         auto      auto  10/100BaseTX
Fa0/24    disabled 100         auto      auto  10/100BaseTX
Gig0/1    disabled 100         auto      auto  10/100BaseTX
Gig0/2    disabled 100         auto      auto  10/100BaseTX

COMLAB_SWITCH#
```

7. Configuration for switch 7 (HR_SWITCH):

- **Purpose:** Acting as an intermediate distribution frame for the HR Department.
- **Configurations Created:**
 - Int fa0/1 configured as a trunk.
 - Int fa0/2 configured as VLAN 20.
 - Int range Fa0/3 - fa0/24, along with interface range G0/1–G0/2, is shut down and assigned to the unused port VLAN.
- **Configuration Screenshots:**

```
HR_SWITCH#sh vlan br
VLAN Name      Status      Ports
-----
1    default      active
10   COMLAB        active
20   HR_DEPT        active      Fa0/2
30   IT_DEPT        active
40   GUEST_WIFI     active
50   SERVERS        active
100  UNUSED_PORTS   active      Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                           Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                           Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                           Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                           Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                           Fa0/23, Fa0/24, Gig0/1, Gig0/2
1002 fddi-default  active
1003 token-ring-default  active
1004 fddinet-default  active
1005 trnet-default   active

HR_SWITCH#
```

```

HR_SWITCH#sh int tr
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     auto      n-802.1q      trunking    1

Port      Vlans allowed on trunk
Fa0/1     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1,10,20,30,40,50,100

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,20,30,40,50,100

HR_SWITCH#

```

```

HR_SWITCH#sh int status
Port      Name      Status      Vlan      Duplex  Speed  Type
Fa0/1     connected trunk      auto      auto    10/100BaseTX
Fa0/2     connected 20        auto      auto    10/100BaseTX
Fa0/3     disabled 100       auto      auto    10/100BaseTX
Fa0/4     disabled 100       auto      auto    10/100BaseTX
Fa0/5     disabled 100       auto      auto    10/100BaseTX
Fa0/6     disabled 100       auto      auto    10/100BaseTX
Fa0/7     disabled 100       auto      auto    10/100BaseTX
Fa0/8     disabled 100       auto      auto    10/100BaseTX
Fa0/9     disabled 100       auto      auto    10/100BaseTX
Fa0/10    disabled 100       auto      auto    10/100BaseTX
Fa0/11    disabled 100       auto      auto    10/100BaseTX
Fa0/12    disabled 100       auto      auto    10/100BaseTX
Fa0/13    disabled 100       auto      auto    10/100BaseTX
Fa0/14    disabled 100       auto      auto    10/100BaseTX
Fa0/15    disabled 100       auto      auto    10/100BaseTX
Fa0/16    disabled 100       auto      auto    10/100BaseTX
Fa0/17    disabled 100       auto      auto    10/100BaseTX
Fa0/18    disabled 100       auto      auto    10/100BaseTX
Fa0/19    disabled 100       auto      auto    10/100BaseTX
Fa0/20    disabled 100       auto      auto    10/100BaseTX
Fa0/21    disabled 100       auto      auto    10/100BaseTX
Fa0/22    disabled 100       auto      auto    10/100BaseTX
Fa0/23    disabled 100       auto      auto    10/100BaseTX
Fa0/24    disabled 100       auto      auto    10/100BaseTX
Gig0/1    disabled 100       auto      auto    10/100BaseTX
Gig0/2    disabled 100       auto      auto    10/100BaseTX

```

```

HR_SWITCH#

```

ACCESS POINT CONFIGURATION

1. Access Point 1 Configuration (Guest Wi-Fi):

- **Purpose:** Acting as a wireless access point for guests.
- **Configurations:**
 - Connected to fa0/2 interface of switch 9 (HELPDESK_SWITCH) as VLAN 40.
 - WPA2-PSK Security.
- **Configuration Screenshot:**

The screenshot shows the 'Access Point1' configuration window with the 'Config' tab selected. The left sidebar has a tree view with 'GLOBAL' and 'INTERFACE' sections. Under 'INTERFACE', 'Port 0' and 'Port 1' are listed, with 'Port 1' selected. The main area displays the configuration for 'Port 1'. The 'Port Status' is 'On'. The 'SSID' is 'Default'. The '2.4 GHz Channel' is '6'. The 'Coverage Range (meters)' is '140.00'. The 'Authentication' section has three radio buttons: 'Disabled', 'WEP', and 'WPA2-PSK', with 'WPA2-PSK' selected. The 'WEP Key' field is empty. The 'PSK Pass Phrase' field contains 'GuestWifi123'. The 'User ID' field is empty. The 'Password' field is empty. The 'Encryption Type' is 'AES'.

Port 1	
Port Status	☒ On
SSID	Default
2.4 GHz Channel	6
Coverage Range (meters)	140.00
Authentication	
☐ Disabled	☐ WEP
☐ WPA-PSK	☒ WPA2-PSK
WEP Key	
PSK Pass Phrase	GuestWifi123
User ID	
Password	
Encryption Type	AES

2. Access Point 2 Configuration (COMLAB):

- **Purpose:** Acting as a wireless access point for the Computer Laboratory Room.
- **Configurations:**
 - Connected via fa0/2 interface of switch 11 (COMLAB_SWITCH) as VLAN 10.
 - WPA2-PSK Security.
- **Configuration Screenshot:**

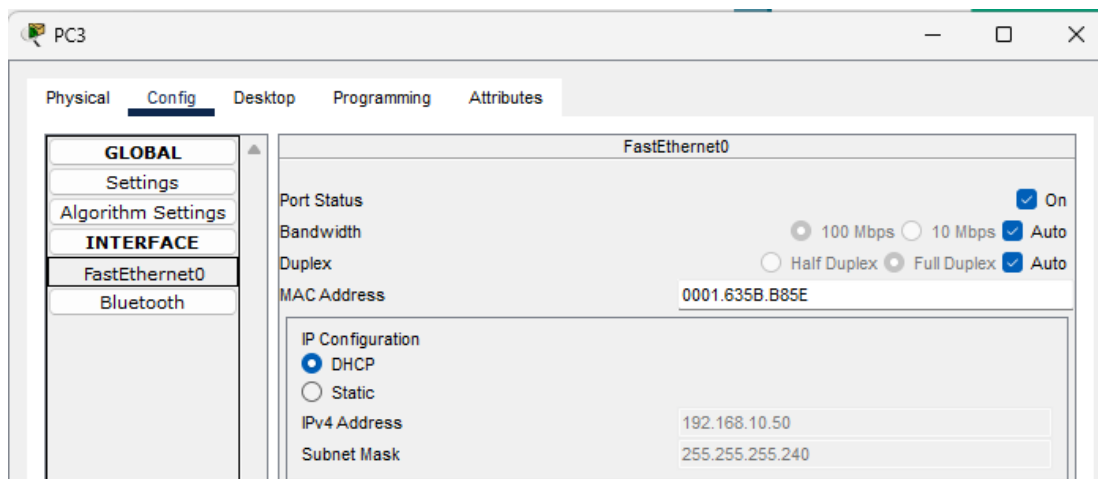
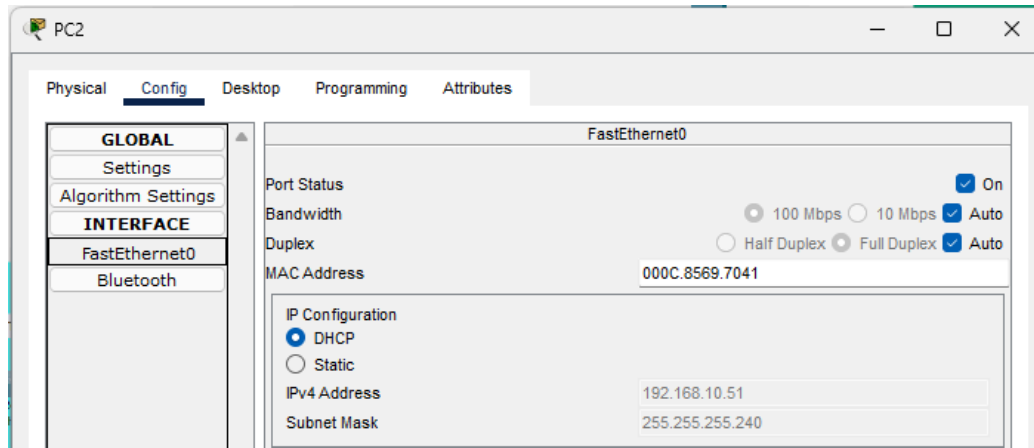
The screenshot shows the 'Access Point2' configuration window with the 'Config' tab selected. The left sidebar has a tree view with 'GLOBAL' and 'INTERFACE' sections. Under 'INTERFACE', 'Port 0' and 'Port 1' are listed, with 'Port 1' currently selected. The main area displays the configuration for 'Port 1'. The 'Port Status' is 'On'. The 'SSID' is 'Default'. The '2.4 GHz Channel' is '6'. The 'Coverage Range (meters)' is '140.00'. Under 'Authentication', 'WPA2-PSK' is selected. The 'WEP Key' field is empty. The 'PSK Pass Phrase' is 'ComLabPass123'. The 'User ID' and 'Password' fields are empty. The 'Encryption Type' is 'AES'. A 'Top' link is at the bottom left.

Port 1	
Port Status	☒ On
SSID	Default
2.4 GHz Channel	6
Coverage Range (meters)	140.00
Authentication	
☐ Disabled	☐ WEP
☐ WPA-PSK	☒ WPA2-PSK
WEP Key	
PSK Pass Phrase	ComLabPass123
User ID	
Password	
Encryption Type	AES

END DEVICES

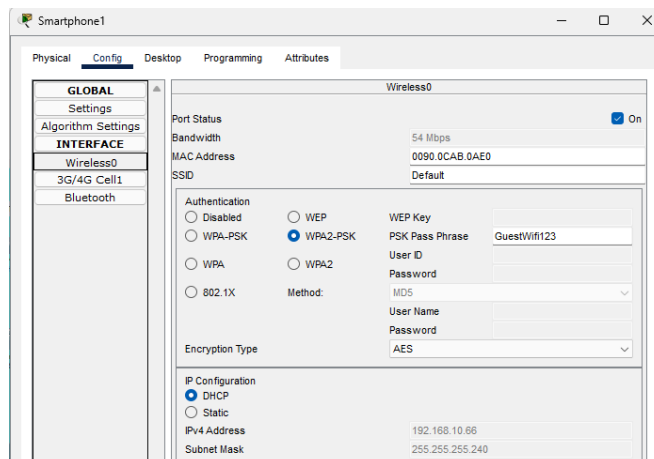
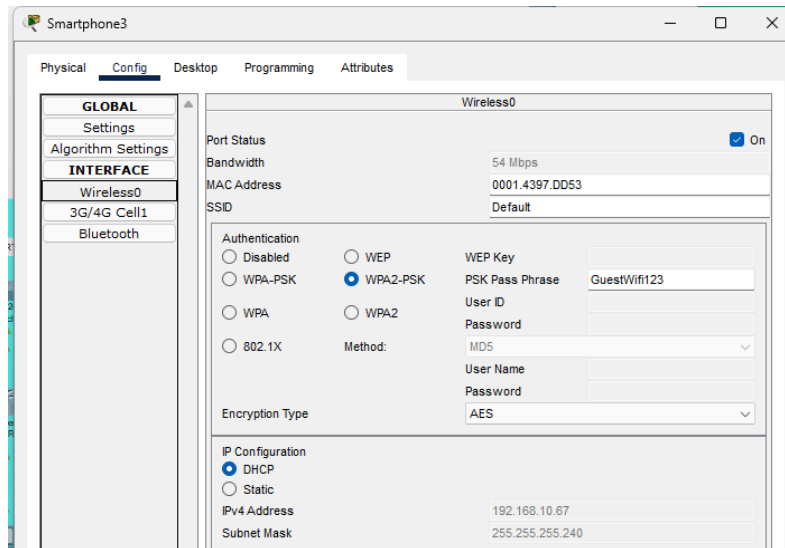
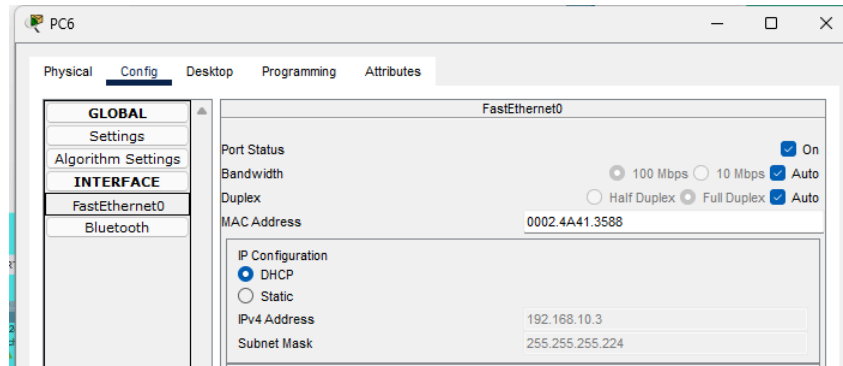
1. IT Department:

- PC2 - Connected via fa0/2 of ITDEPT_SWITCH.
- PC3 - Connected via fa0/3 of ITDEPT_SWITCH.
- **Configuration Screenshots:**



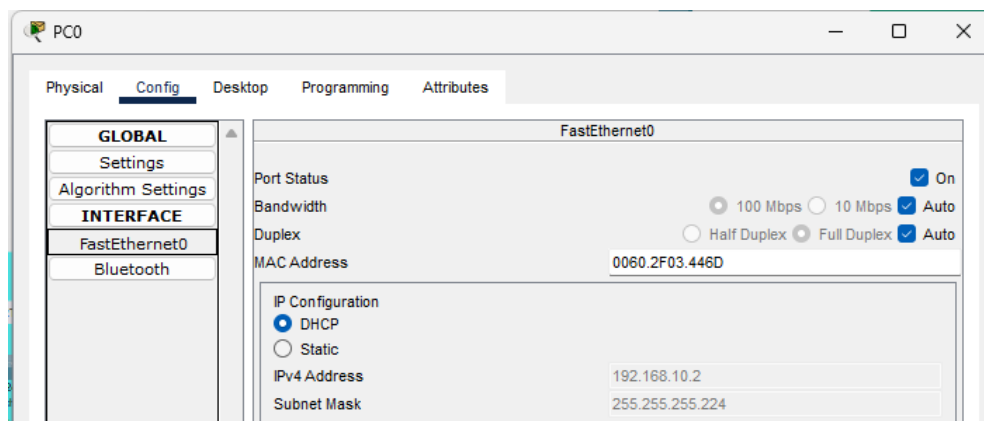
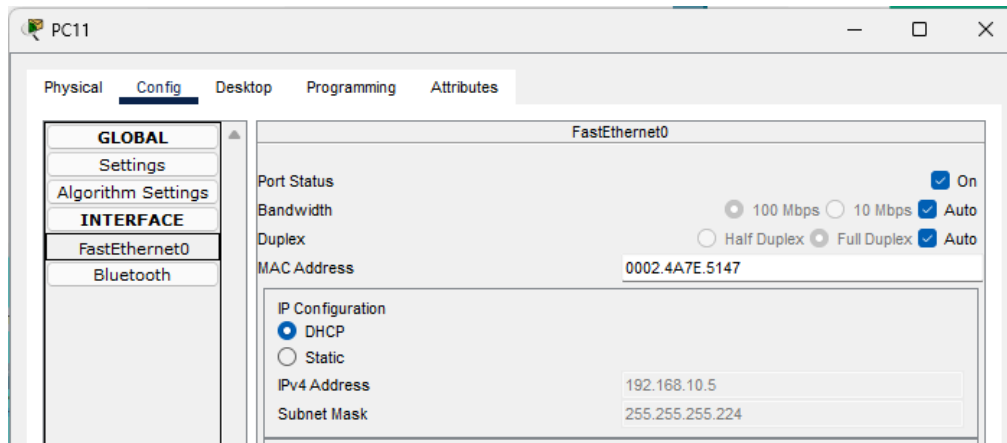
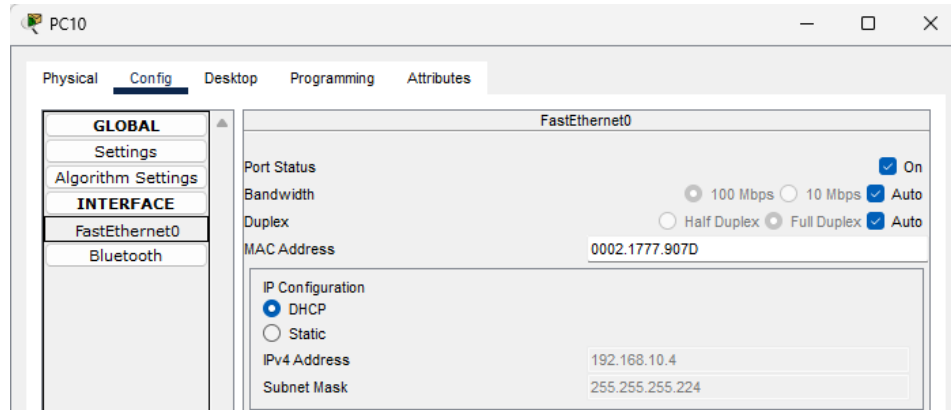
2. Physical Helpdesk Room:

- PC6 - Connected via fa0/3 of HELPDESK_SWITCH.
- Smartphone 1 & 3 - Connected wirelessly via access point 1.
- **Configuration Screenshots:**



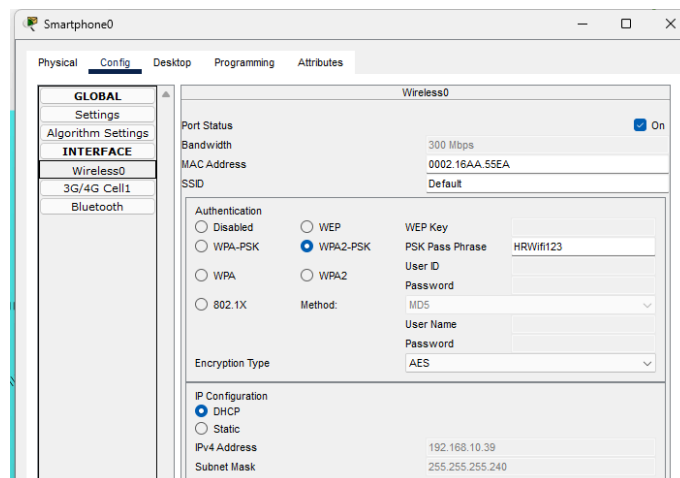
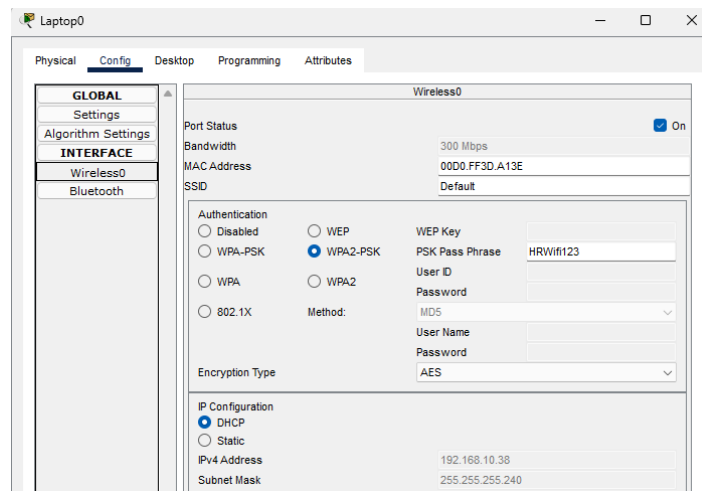
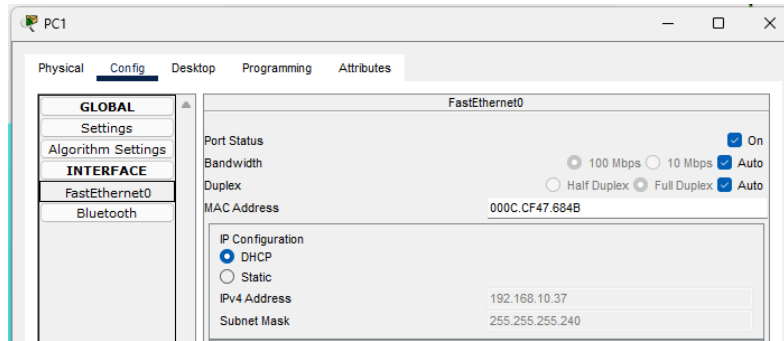
3. Computer Laboratory Room:

- PC10 - Connected via fa0/3 of COMLAB_SWITCH.
- PC11 - Connected via fa0/4 of COMLAB_SWITCH.
- PC0 - Connected via fa0/5 of COMLAB_SWITCH.
- Smartphone 2 - Connected wirelessly via access point 2.
- **Configuration Screenshots:**



4. HR Department:

- PC1 - Connected via g0/1 of HOME ROUTER PT-AC.
- Laptop 0 - Connected wirelessly via HOME ROUTER PT-AC.
- Smartphone 0 - Connected wirelessly via HOME ROUTER PT-AC.
- **Configuration Screenshots:**

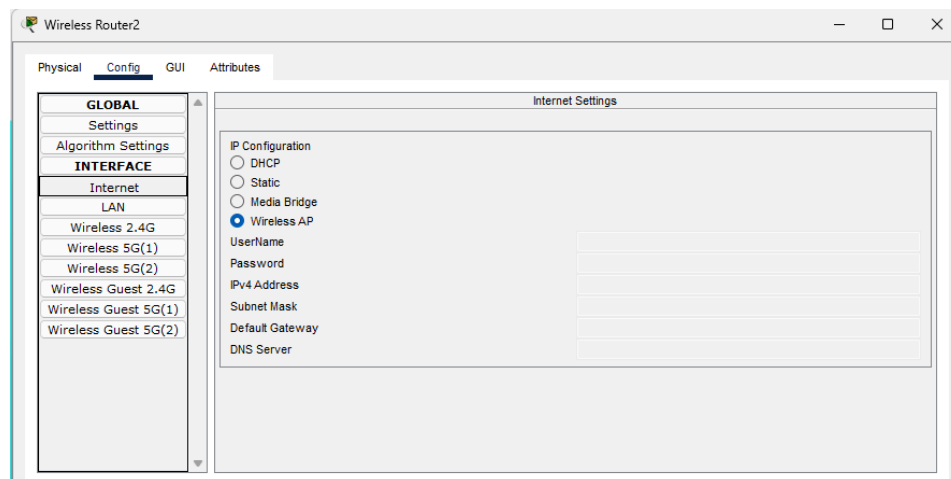


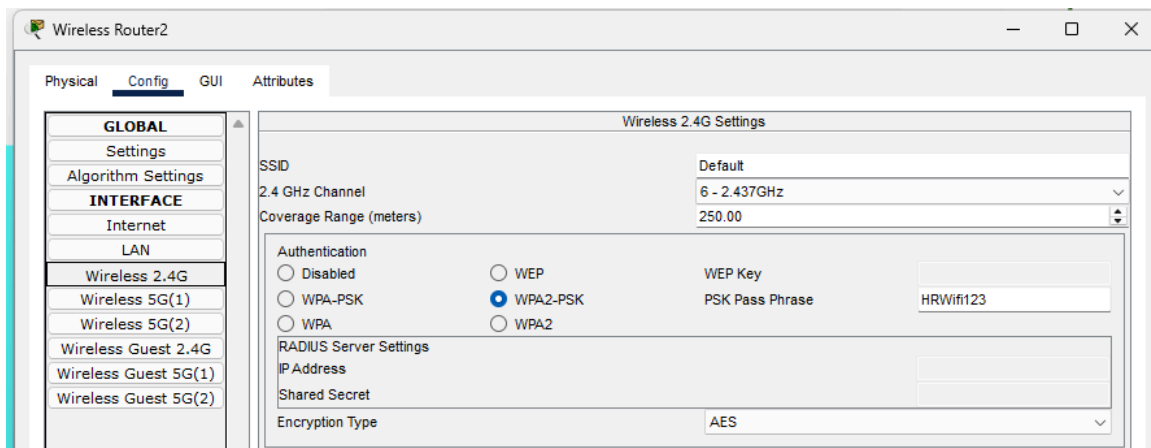
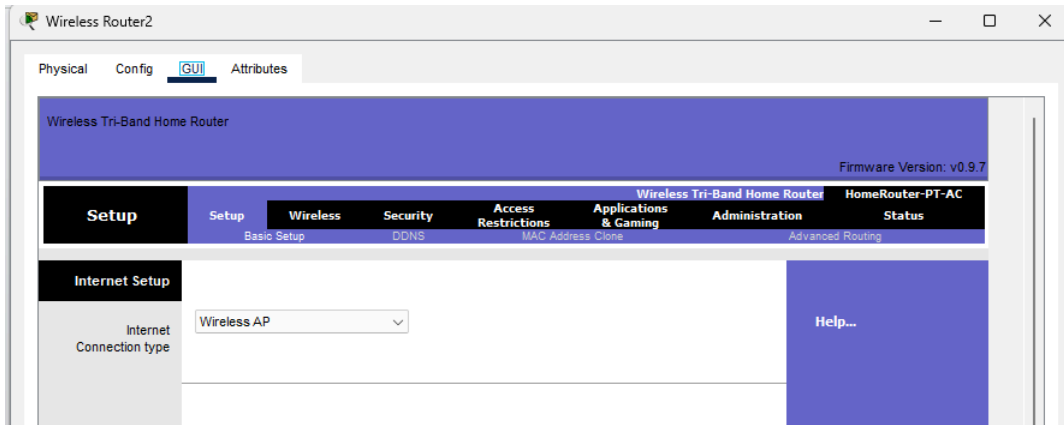
HOME ROUTER (PT-AC) CONFIGURATION

By default, home routers have their own internal configuration for their default IP address and their own DHCP pool. Since the goal of this laboratory is to showcase the DHCP Relay, the home router must be configured to become a working access point instead of working as a full home router.

Configurations made:

- Turned off the DHCP server and set the home router as a wireless access point.
- Instead of plugging the fa0/2 port of the HR_SWITCH directly into the WAN port of the home router PT-AC, it was plugged into the G0/2 port of the home router.
 - **Explanation:** WAN ports and LAN ports have different MAC addresses based on their network interface card (NIC).
 - **Security Mitigation:** We want to make sure that the MAC address of the LAN port is the one being learned by our HR_SWITCH, not the WAN port.
- **Configuration Screenshot:**





BREAK/FIX SCENARIO

Scenario: Provisioning of home router as an AP to the HR Department

Identification: Newly bought unconfigured home router.

Root Cause: A consumer router (PT-AC) placed in HR was running active DHCP/NAT out of the box, conflicting with Server0.

Further explanations with provided screenshots:

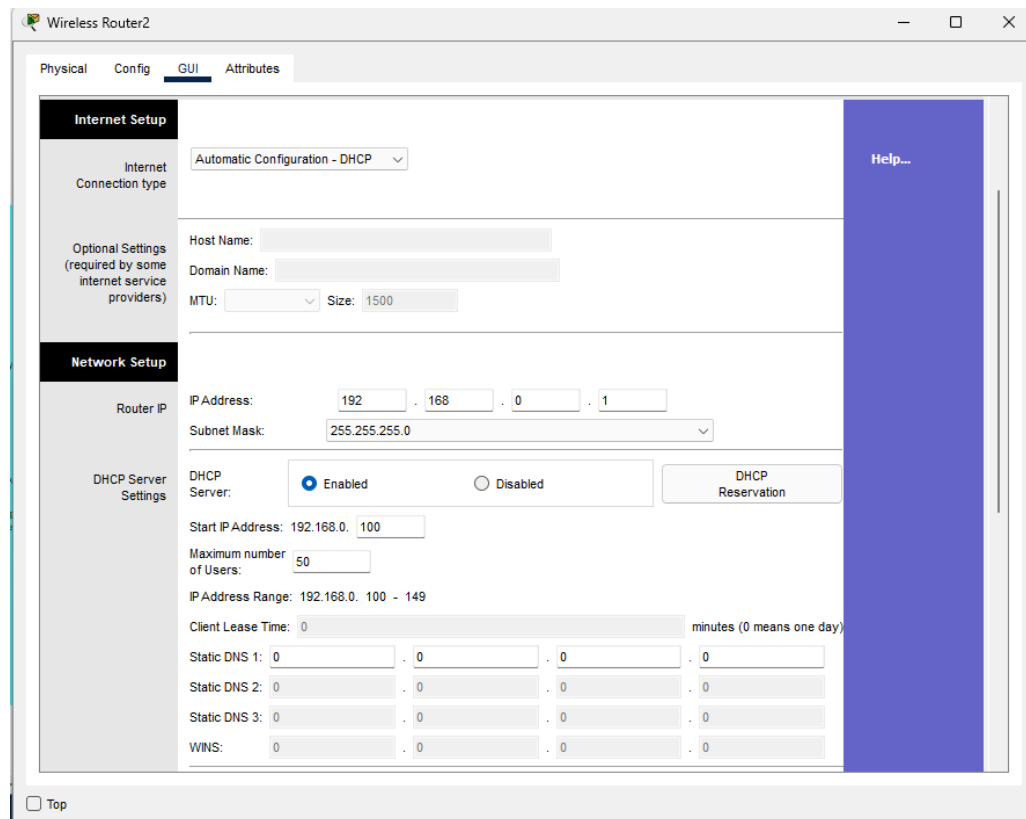


Figure 2: Unconfigured Home Router

Based on the given screenshot, the home router has an IP address of 192.168.0.1 with a subnet of /24. It also has an active pre-configured DHCP Server resulting in conflicts with the centralized server in the network topology.

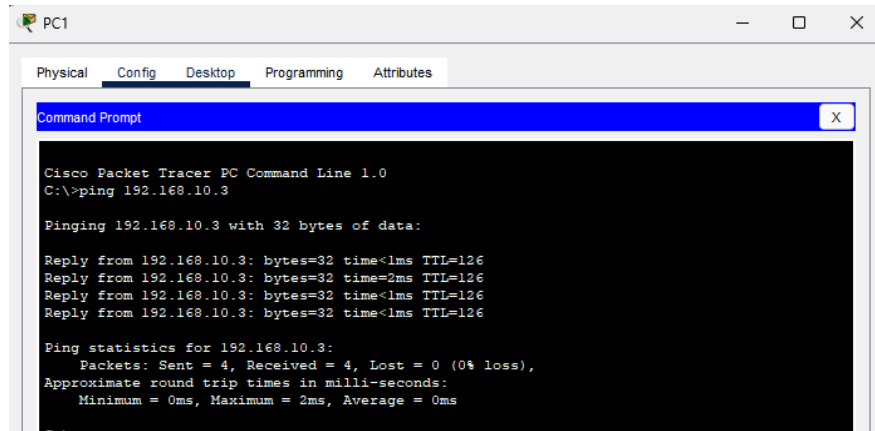


Figure 3: PC10 of VLAN 10 reachable by PC1 inside the home router

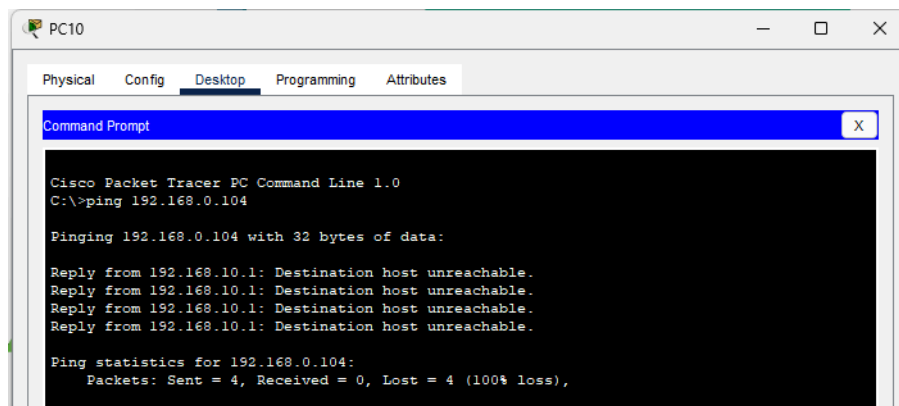


Figure 4: PC10 of VLAN 10 cannot reach PC1 inside the home router

Additionally, PC1 inside the home router can reach PC10 inside VLAN 10 without a problem. On the other hand, PC10 inside VLAN 10 cannot reach PC1 inside the home router, possibly due to its built-in firewall denying outside traffic and its own NAT.

To address the issue, the following fixes must be applied:

Fix Applied: Disabled local DHCP/NAT on the home router, bypassed the WAN port, connected via LAN port G0/2 to convert it into a pure Layer 2 Access Point.